

MSBA 265: Business Analytics Topics

Foundational Module 1: Practical Homework Assignment

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Part 1: Workspace Setup, Git Ingestion & Procurement

1.1 Workspace Setup

The project workspace was initialized with Git and a Python virtual environment, with a `.gitignore` configured for local environment and cache files.

1.2 Data Ingestion

A Python ingestion script, `data/download_data.py`, was created to programmatically download the French Motor Third Party Liability Claims (`freMTPL2freq.csv`) dataset from OpenML and save it as `data/raw_business_data.csv`. The resulting raw dataset contains **678,013** records and **12** features.

1.3 Git Version Control

The project setup and ingestion artifacts were committed and pushed to the public GitHub repository.

Part 2: Raw Boundary Audits & Business Data Dictionary

2.2 Raw Boundary Audit

The raw dataset was audited for minimum and maximum values of the required features.

Feature	Minimum	Maximum
DrivAge	18	100
Exposure	0.002732	2.01
BonusMalus	50	230

The `DrivAge` range contains no negative ages or child policyholder values. `Exposure` contains no negative durations, and the minimum value is approximately 0.0027 years. `BonusMalus` values remain within the observed range of the raw dataset.

The exported Business Data Dictionary is available as `reports/data_dictionary.csv`.

2.4 Exposure Offset & Density/Area Defense

Exposure works as a rate offset because not every policy in the dataset was active for the same amount of time.

For example, a policy with an Exposure of 1.0 was active for a full year, while a policy with an Exposure of 0.25 was only active for about three months. A claim from these two policies should not be treated as representing the same claim rate, so `log(Exposure)` is used as an offset to account for the amount of time each policy was actually at risk.

Density and Area should also not both be used in the same `linear regression` model because both contain geographic risk information. Density gives a continuous measure of how populated an area is, while Area divides locations into categories. Since these variables can contain overlapping information about location, including both can create multicollinearity and make the individual effects of the variables harder to interpret.

Part 3: Overriding Automated Rules & Correlation Visuals

3.1 Pearson Correlation Heatmap

The Pearson correlation heatmap shows the linear relationships among the numerical features.

3.2 ClaimNb Transformation Defense

The automated diagnostics recommend a log transformation for `ClaimNb` because it is heavily right-skewed, but we are going to override that recommendation.

More than 95% of the policies have zero claims, and `log(0)` is not defined. Also, `ClaimNb` is a count of actual claim events, not a continuous measurement.

Because of this, it makes more sense to keep `ClaimNb` in its original count form and use a count-based model such as Poisson. `log(Exposure)` can then be used as an offset so the model accounts for how long each policy was actually active.



Figure 1: Pearson Correlation Heatmap

Part 4: Distribution Audits & Written Plot Statements

4.1 Feature Distributions

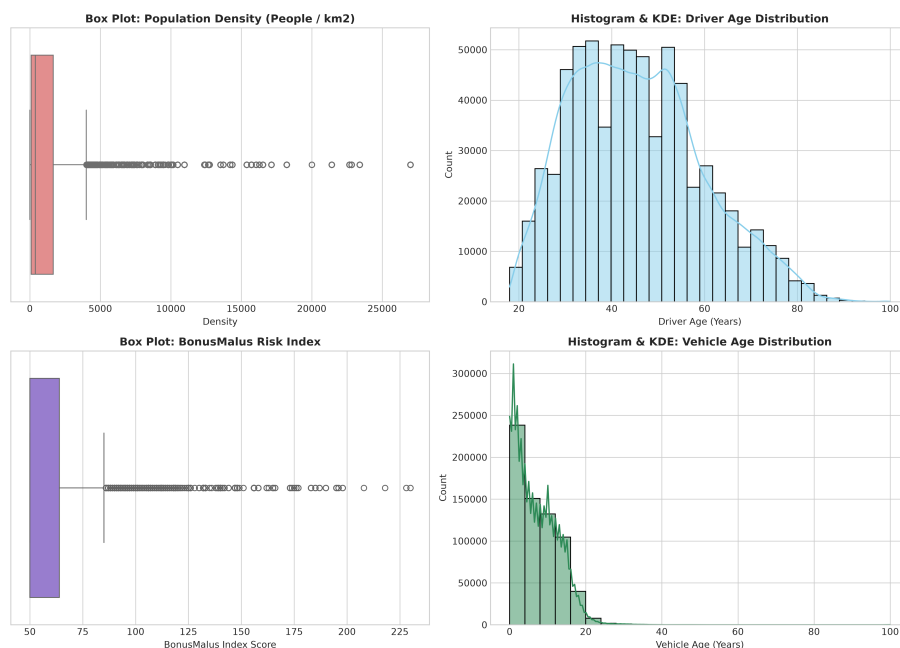


Figure 2: Feature Distributions

4.2 Distribution Interpretations

Density Box Plot

- **Distribution Shape:** Extremely right-skewed ($\text{skew} = 4.65$), with most values at lower densities and a long tail toward highly populated areas.
- **Key Metrics/Whiskers:** Median = 393, $Q1 = 92$, $Q3 = 1,658$. The upper Tukey fence is 4,007, but higher values can still represent valid metropolitan areas.
- **Recommended Analytics Action:** Apply $\log(1 + \text{Density})$ before using Density in linear regression.

DrivAge Histogram & KDE

- **Distribution Shape:** Fairly symmetric and close to a bell-shaped distribution ($\text{skew} = 0.44$).
- **Key Metrics/Whiskers:** Median = 44, $Q1 = 34$, and $Q3 = 55$.

- **Recommended Analytics Action:** Standard Z-score scaling can be used, with no major outlier removal needed.

BonusMalus Box Plot

- **Distribution Shape:** Right-skewed ($\text{skew} = 1.73$), with most values concentrated toward the lower end.
- **Key Metrics/Whiskers:** Median = 50, Q1 = 50, Q3 = 64, with an upper Tukey fence of 85. Values above 100 represent drivers receiving a surcharge.
- **Recommended Analytics Action:** Keep it as a continuous variable and create an `IsSurcharge` indicator for `BonusMalus > 100`.

VehAge Histogram & KDE

- **Distribution Shape:** Moderately right-skewed ($\text{skew} = 1.15$), with more newer vehicles than older ones.
- **Key Metrics/Whiskers:** Median = 6, Q1 = 2, and Q3 = 11. The distribution is concentrated around younger vehicle ages.
- **Recommended Analytics Action:** Bin into groups such as 0–5, 6–10, and 11+, or apply scaling.

Correlation Heatmap

- **Distribution Shape:** The numerical features generally show weak linear relationships with each other.
- **Key Metrics/Whiskers:** The strongest non-self correlation is between `DrivAge` and `BonusMalus` at $r = -0.48$. No feature pair has $|r| \geq 0.85$.
- **Recommended Analytics Action:** Check correlated predictors before regression. Pearson correlation only captures linear relationships, so near-zero correlation does not rule out nonlinear relationships.

Part 5: Production Pipeline Scripting & Final PDF Export

5.1 Production Outlier Pipeline

The production outlier filtering script `src/clean_outliers.py` was executed from the project root using:

```
python src/clean_outliers.py
```

The script applies Tukey's $1.5 \times \text{IQR}$ rule to the `Density` feature.

Metric	Result
Initial Dataset Records	678,013

Metric	Result
Records Removed	77,566
Percentage Removed	11.44%
Final Cleaned Records	600,447

The cleaned dataset was generated at `data/cleaned_business_data.csv`.

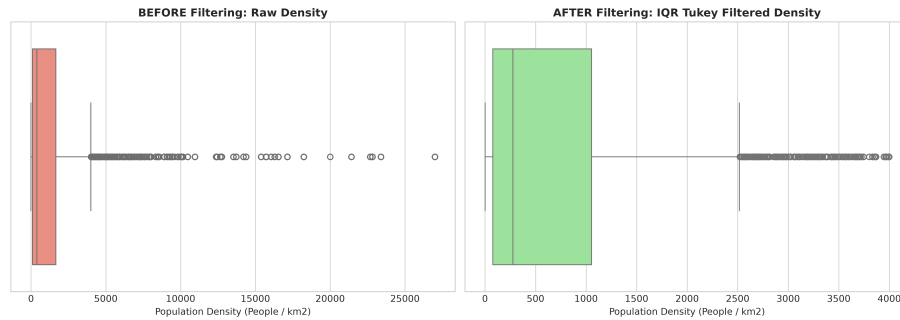


Figure 3: Outlier Filtering Comparison

5.2 Final Report

This document contains the final written answers and required plot figures for the Module 1 Practical Homework Assignment. It is compiled into `Module1 Homework Report.pdf` for submission. ## 5.3 GitHub Verification All required project artifacts are tracked in Git and pushed to the public GitHub repository.

The repository contains the required source scripts, notebook, raw and cleaned datasets, data dictionary, generated figures, README reproduction instructions, and final PDF report.

Reproduction

From the project root, create and activate the virtual environment, install the required dependencies, and execute the data ingestion and production pipeline scripts.

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
python data/download_data.py
python src/clean_outliers.py
```

The exploratory analysis and data dictionary can be reproduced by opening `notebooks/01_eda_and_data_dictionary.ipynb`.

The generated artifacts are stored in the `data/` and `reports/` directories.

Project Artifacts

- `README.md`
- `requirements.txt`
- `data/download_data.py`
- `data/raw_business_data.csv`
- `data/cleaned_business_data.csv`
- `notebooks/01_eda_and_data_dictionary.ipynb`
- `src/clean_outliers.py`
- `reports/data_dictionary.csv`
- `reports/figures/feature_distributions.png`
- `reports/figures/correlation_heatmap.png`
- `reports/figures/outlier_filtering_comparison.png`
- `Module1 Homework Report.pdf`